

# Twisted States in Higgs Boson Decay into Fermion Antifermion Pair

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# 1 Introduction

## Problem Statement

Consider the decay of the Higgs boson into a pair of fermions  $f\bar{f}$ . The quantum state of the fermions is given by:

$$|f\bar{f}\rangle = \int \frac{d\vec{p}_1}{(2\pi)^3\sqrt{2E_1}} \frac{d\vec{p}_2}{(2\pi)^3\sqrt{2E_2}} \psi(\vec{p}_1, s_1; \vec{p}_2, s_2) |f(\vec{p}_1, s_1), \bar{f}(\vec{p}_2, s_2)\rangle,$$

where  $\psi$  is the two-particle wave function of fermions in momentum representation,  $s_i$  are the spin 4-vectors. In the rest frame of the fermion  $s = (0, \boldsymbol{\xi})$ .

1. Find  $\psi$  using the matrix element:

$$\langle f(\vec{p}_1, s_1), \bar{f}(\vec{p}_2, s_2) | S - 1 | \text{Higgs boson} \rangle.$$

2. Now one can calculate the wave function in coordinate representation:

$$\psi(\vec{x}_1, \vec{x}_2) = \int \frac{d\vec{p}_1}{(2\pi)^3} \frac{d\vec{p}_2}{(2\pi)^3} \tilde{\psi}(\vec{p}_1, s_1; \vec{p}_2, s_2) e^{+i(\vec{p}_1\vec{x}_1 + \vec{p}_2\vec{x}_2)}.$$

Show that  $\psi(\vec{x}_1, \vec{x}_2)$  depends on the coordinate difference  $\vec{r} = \vec{x}_1 - \vec{x}_2$  and calculate the wave function  $\psi(\vec{r})$ .

3. Expand the wave function in a series of spherical harmonics and investigate their behavior as a function of  $\boldsymbol{\xi}_1 \cdot \hat{\vec{r}}, \boldsymbol{\xi}_2 \cdot \hat{\vec{r}}, \boldsymbol{\xi}_1 \cdot \boldsymbol{\xi}_2$ . Give an interpretation.

## 2 Find wavefunction

Using the Fourier transform

$$\begin{aligned} \psi(\vec{r}_1, \vec{r}_2) &= \int \frac{d^3p_1 d^3p_2}{(2\pi)^6 \sqrt{2E_1 2E_2}} \tilde{\psi}(\vec{p}_1, \vec{p}_2) e^{i(\dots)} \\ &= \int \frac{d^2p_1 d^2p_2}{(2\pi)^3} \langle f\bar{f} | S - 1 \rangle e^{i(\dots)} \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \frac{m_f}{v} \bar{u}(\vec{p}_1, \vec{s}_1) u(-\vec{p}_2, \vec{s}_2) e^{i(\dots)} \\ S | i \rangle &= | f \rangle = \dots + \int \frac{d^3p_1 d^3p_2}{(2\pi)^6} \frac{\tilde{\psi}(\vec{p}_1, \vec{p}_2)}{\sqrt{2E_1 2E_2}} | \vec{p}_1, \vec{s}_1; \vec{p}_2, \vec{s}_2 \rangle + \dots \end{aligned}$$

$$\langle \vec{p}_1 \vec{p}_2 | \vec{p}_1 \vec{p}_2 \rangle = (2\pi)^3 \sqrt{2E_1 2E_2} [\delta^3(\vec{p}_1 - \vec{p}_2) \delta'(\vec{p}_2 - \vec{p}_1') - \delta^3(\vec{p}_1 - \vec{p}_2) \delta'(\vec{p}_2 - \vec{p}_1)]$$

Momentum

$$\begin{aligned} |p\rangle &\equiv \sqrt{2E_p} a_p^\dagger |0\rangle \\ \langle \vec{q} | \vec{p} \rangle &= \langle \vec{q} | a_q \sqrt{2E_q} \sqrt{2E_p} a_p^\dagger |0\rangle = 2E_p (2\pi)^3 \delta^3(\vec{q} - \vec{p}) \\ a_q a_p^\dagger &= -a_q^\dagger a_p + \underbrace{\{a_q, a_p^\dagger\}}_{(2\pi^3)\delta^3(\vec{q}-\vec{p})} \end{aligned}$$

We'll get the following norm

$$\langle \vec{q}, s | \vec{p}, r \rangle = 2E_p (2\pi)^3 \delta^3(\vec{q} - \vec{p}) \delta_{sr}$$

Substitute in our decay

$$\begin{aligned} \langle q_1, r_1; q_2, r_2 | p_1 s_1; p_2 s_2 \rangle &= \sqrt{2E_{p_1} 2E_{p_2} 2E_{q_1} 2E_{q_2}} \langle 0 | a_{q_1 r_1} a_{q_2 r_2} a_{p_1 s_1}^\dagger a_{p_2 s_2}^\dagger | 0 \rangle \\ &= \sqrt{2E_{p_1} 2E_{p_2} 2E_{q_1} 2E_{q_2}} \langle 0 | a_{q_1 r_1} (-a_{q_2 r_2}^\dagger a_{p_1 s_1} + \{a_{q_2 r_2}, a_{p_1 s_1}^\dagger\}) a_{p_2 s_2}^\dagger | 0 \rangle \\ &= \{a_{q_2 r_2}, a_{p_1 s_1}^\dagger\} \langle 0 | a_{q_1 r_1} a_{p_2 s_2}^\dagger | 0 \rangle - \langle 0 | a_{q_1 r_1} a_{q_2 r_2}^\dagger a_{p_1 s_1} a_{p_2 s_2}^\dagger | 0 \rangle \\ &= \{a_{q_2 r_2}, a_{p_1 s_1}^\dagger\} \{a_{q_1 r_1}, a_{p_2 s_2}^\dagger\} - \{a_{q_1 r_1}, a_{q_2 r_2}^\dagger\} \{a_{p_1 s_1}, a_{p_2 s_2}^\dagger\} \end{aligned}$$

Finally will get

$$\begin{aligned} \langle \vec{q}_1 r_1; \vec{q}_2 r_2 | S | H \rangle &= \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6 \sqrt{2E_{p_1} 2E_{p_2}}} \tilde{\psi}(\vec{p}_1, \vec{p}_2) \langle q_1 r_1; q_2 r_2 | p_1 s_1; p_2 s_2 \rangle \\ &= \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6 \sqrt{2E_{p_1} 2E_{p_2}}} \tilde{\psi}(\vec{p}_1, \vec{p}_2) [(2\pi)^6 \delta^3(q_2 - p_1) \delta_{r_2 s_1} \delta^3(q_1 - p_1) \delta_{r_2 s_1} - (2\pi)^6 \delta^3(q_2 - p_2)] \\ &= \frac{\tilde{\psi}(q_2, q_1)}{\sqrt{2E_{q_1} 2E_{q_2}}} \delta_{r_2 s_1} \delta_{r_1 s_2} - \frac{\tilde{\psi}(q_1, q_2)}{\sqrt{2E_{q_1} 2E_{q_2}}} \delta_{r_1 s_1} \delta_{r_2 s_2} \end{aligned}$$

??

$$\frac{\tilde{\psi}(q_2 r_2; q_1 r_1) - \tilde{\psi}(q_1 r_1; q_2 r_2)}{\sqrt{2E_{q_1} 2E_{q_2}}}$$

$$\tilde{\psi}(q_1 r_1; q_2 r_2) \sim \sqrt{2E_{q_1} 2E_{q_2}} \langle q_1 r_1; q_2 r_2 | S - 1 | H \rangle$$

$$\begin{aligned} \tilde{\psi}(\vec{r}_1 s_1; \vec{r}_2 s_2) &= \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6 \sqrt{2E_{p_1} 2E_{p_2}}} \tilde{\psi} e^{i\vec{p}_1 \cdot \vec{r}_1 + i\vec{p}_2 \cdot \vec{r}_2} \\ &= \int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6} \underbrace{\langle p_1 s_1; p_2 s_2 | S - 1 | H \rangle}_{(2\pi)^4 \delta^4(P_H - P_1 - P_2) i \frac{m_f}{v} \bar{u}(p_1 s_1) v(p_2 s_2)} e^{i\vec{p}_1 \cdot \vec{r}_1 + i\vec{p}_2 \cdot \vec{r}_2} \end{aligned}$$

Using the spherical system of coordinates

$$\int \frac{d^3 p_1 d^3 p_2}{(2\pi)^6} \delta^4(P_H - E_{p_1} - E_{p_2}) i \frac{m_f}{v} \bar{u}(p_1 s_1) v(p_2 s_2) \Big|_{\vec{p}_2 = -\vec{p}_1} e^{i\vec{p}_1 \cdot \vec{r}_1 + i\vec{p}_1 \cdot \vec{r}_2}$$

$$d^3 P_1 = \underbrace{P_1^2 dP_1}_{p_1 E_1 dE_1} d\Omega_{p_1}$$

$$E^2 = p^2 + m^2, \quad EdE = p dp$$

$$p_1^2 dp_1 = p_1 p_1 dp_1 = p_1 E_1 dE_1 = \sqrt{E_1^2 - m^2} E_1 dE_1$$

Using the property of delta function:  $\int f(x) \delta(g(x)) = \sum_i \frac{f(x_i)}{|g'(x_i)|}$  and  $g \equiv m_H - 2E_1 \equiv g(E_1)$ ,  $g'(E_1) = -2$

$$\begin{aligned} & \frac{1}{(2\pi)^6} \int d\Omega dE_1 \sqrt{E_1^2 - m^2} \delta^4(m_H - 2E_1) i \frac{m_f}{v} \bar{u}(\vec{p}_1, s_1) v(-\vec{p}_1, s_2) \\ &= \frac{1}{(2\pi)^6} \frac{1}{2} \int d\Omega \sqrt{\left(\frac{m_H}{2}\right)^2 - m^2} \frac{m_H}{2} i \frac{m_f}{v} \bar{u}(\vec{p}_1, s_1) v(-\vec{p}_1, s_2) \Big|_{|\vec{p}_1| = \sqrt{\left(\frac{m_H}{2}\right)^2 - m^2}} \end{aligned}$$

We have reached this expression

$$\psi(\vec{r}_1 - \vec{r}_2; s_1, s_2) \sim \int d\theta \sin \theta d\phi \underbrace{f(\theta, \phi)}_{\bar{u}v} e^{i|\vec{p}| \underbrace{(\vec{r}_1 - \vec{r}_2) \cdot \hat{n}}_{\equiv \vec{r}}}$$

where  $\hat{n} = \frac{\vec{p}}{|\vec{p}|}$

### 3 Fermion Antifermion product

From Dirac's equation for fermion

$$u(p) = \begin{pmatrix} \sqrt{E_p + m_f} \chi(\vec{s}_1) \\ \sqrt{E_p - m_f} (\vec{\sigma} \cdot \hat{n}_p) \chi(\vec{s}_1) \end{pmatrix}$$

and antifermion  $v(p, s) = c\gamma^0 u^*(p, s)$

$$v(-p) = \begin{pmatrix} \sqrt{E_p - m_f} (\vec{\sigma} \cdot \hat{n}_{-p}) \eta(\vec{s}_2) \\ \sqrt{E_p + m_f} \eta(\vec{s}_2) \end{pmatrix}$$

taking in consideration  $\hat{n}_{-p} = -\hat{n}_p$  and  $\eta \hat{s}_2 = i\sigma_y \chi^*(\hat{s}_2)$

And for

$$\chi(\vec{s}_1) = \begin{pmatrix} \cos \frac{\theta_1}{2} \\ \sin \frac{\theta_1}{2} e^{i\phi_1} \end{pmatrix}$$

we get (keeping in mind that  $\mathbf{p}_1 \uparrow\uparrow \mathbf{s}_1$  and  $\mathbf{p}_2 \uparrow\downarrow \mathbf{s}_2$  we get the following substitution  $(\theta_2, \phi_2) \rightarrow (\pi - \theta_2, \phi_2 + \pi)$ )

$$\eta(\mathbf{s}_2) = i\sigma_y \chi^*(\mathbf{s}_2) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} \cos \frac{\pi - \theta_2}{2} \\ \sin \frac{\pi - \theta_2}{2} e^{-i(\phi_2 + \pi)} \end{pmatrix} = \begin{pmatrix} -\cos \frac{\theta_2}{2} e^{-i\phi_2} \\ -\sin \frac{\theta_2}{2} \end{pmatrix}$$

Substitute in our

$$\bar{u} = u^\dagger \gamma^0 = (\sqrt{E_p + m_f} \chi^\dagger, -\sqrt{E_p - m_f} \chi^\dagger (\vec{\sigma} \cdot \hat{n}_p))$$

$$v = \begin{pmatrix} \sqrt{E_p - m_f} (\vec{\sigma} \cdot \hat{n}_{-p}) \eta(\vec{s}_2) \\ \sqrt{E_p + m_f} \eta(\vec{s}_2) \end{pmatrix} = \begin{pmatrix} -\sqrt{E_p - m_f} (\vec{\sigma} \cdot \hat{n}_p) \eta(\vec{s}_2) \\ \sqrt{E_p + m_f} \eta(\vec{s}_2) \end{pmatrix}$$

$$\bar{u}v = -2\sqrt{(E_p + m_f)(E_p - m_f)} \chi^\dagger (\vec{\sigma} \cdot \hat{n}_p) \eta$$

$$\chi^\dagger(\mathbf{s}_1) = (\cos \frac{\theta_1}{2}, \sin \frac{\theta_1}{2} e^{-i\phi_1})$$

$$\eta(\mathbf{s}_2) = \begin{pmatrix} -\cos \frac{\theta_2}{2} e^{-i\phi_2} \\ -\sin \frac{\theta_2}{2} \end{pmatrix}$$

Expression for

$$\vec{\sigma} \cdot \hat{n}_p = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z = \begin{pmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{pmatrix}$$

$$\hat{n}_p = (\sin \theta_p \cos \phi_p, \sin \theta_p \sin \phi_p, \cos \theta_p)$$

$$\vec{s}_i = (\sin \theta_i \cos \phi_i, \sin \theta_i \sin \phi_i, \cos \theta_i)$$

$$\vec{\sigma} \cdot \hat{n}_p = \begin{pmatrix} \cos \theta_p & \sin \theta_p e^{-i\phi_p} \\ \sin \theta_p e^{i\phi_p} & -\cos \theta_p \end{pmatrix} = \begin{pmatrix} \cos \theta_p & \sin \theta_p e^{-i\phi_p} \\ \sin \theta_p e^{i\phi_p} & -\cos \theta_p \end{pmatrix}$$

$$(\vec{\sigma} \cdot \hat{n}_p) \eta(\mathbf{s}_2) = \begin{pmatrix} \cos \theta_p & \sin \theta_p e^{-i\phi_p} \\ \sin \theta_p e^{i\phi_p} & -\cos \theta_p \end{pmatrix} \begin{pmatrix} -\cos \frac{\theta_2}{2} e^{-i\phi_2} \\ -\sin \frac{\theta_2}{2} \end{pmatrix} = -\cos \frac{\theta_2}{2} e^{-i\phi_2} \begin{pmatrix} \cos \theta_p \\ \sin \theta_p e^{i\phi_p} \end{pmatrix} - \sin \frac{\theta_2}{2} \begin{pmatrix} \sin \theta_p e^{-i\phi_p} \\ -\cos \theta_p \end{pmatrix}$$

$$\begin{aligned}
\chi^\dagger(\vec{s}_1) (\hat{n}_p \cdot \boldsymbol{\sigma}) \eta(\vec{s}_2) &= \left[ \cos \frac{\theta_1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sin \frac{\theta_1}{2} e^{-i\phi_1} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] \\
&\times \left[ -\cos \frac{\theta_2}{2} e^{-i\phi_2} \begin{pmatrix} \cos \theta_p \\ \sin \theta_p e^{i\phi_p} \end{pmatrix} - \sin \frac{\theta_2}{2} \begin{pmatrix} \sin \theta_p e^{-i\phi_p} \\ -\cos \theta_p \end{pmatrix} \right]^T \\
&= -\cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_2} \cos \theta_p - \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} \sin \theta_p e^{-i\phi_p} \\
&\quad - \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i(\phi_1+\phi_2)} \sin \theta_p e^{i\phi_p} + \sin \frac{\theta_1}{2} e^{-i\phi_1} \sin \frac{\theta_2}{2} \cos \theta_p
\end{aligned}$$

## 4 Calculating integral in spherical coordinates

$$\begin{aligned}
\int d\hat{n} \chi^\dagger(\vec{s}_1) (\hat{n} \cdot \boldsymbol{\sigma}) \eta(\vec{s}_2) &= \int_0^{2\pi} d\phi_p \int_0^\pi d\theta_p \sin \theta_p e^{i\vec{p} \cdot \vec{r}} \times \\
&\left[ -\cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_2} \cos \theta_p - \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} \sin \theta_p e^{-i\phi_p} \right. \\
&\quad \left. - \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i(\phi_1+\phi_2)} \sin \theta_p e^{i\phi_p} + \sin \frac{\theta_1}{2} e^{-i\phi_1} \sin \frac{\theta_2}{2} \cos \theta_p \right]
\end{aligned}$$

Some expressions for spherical harmonics

$$Y_0^0 = \frac{1}{2} \sqrt{\frac{1}{\pi}} \quad Y_1^1 = -\frac{1}{2} \sqrt{\frac{3}{2\pi}} \sin \theta e^{i\phi} \quad (1)$$

$$Y_1^0 = \frac{1}{2} \sqrt{\frac{3}{\pi}} \cos \theta \quad Y_1^{-1} = \frac{1}{2} \sqrt{\frac{3}{2\pi}} \sin \theta e^{-i\phi} \quad (2)$$

$$Y_\ell^{m*}(\theta, \varphi) = (-1)^m Y_\ell^{-m}(\theta, \varphi) \quad (3)$$

Using the expression for  $e^{i\vec{p} \cdot \vec{r}}$  and substituting a trigonometric function for the corresponding spherical harmonic

$$\begin{aligned}
&\int_0^{2\pi} d\phi_p \int_0^\pi d\theta_p \sin \theta_p e^{i\vec{p} \cdot \vec{r}} \sum_{l=0}^{\infty} \sum_{m=-l}^l 4\pi i^l j_l(kr) Y_l^m(\theta_p, \varphi_p) Y_l^{m*}(\theta_r, \varphi_r) \times \\
&\left[ -\cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_2} \cdot 2 \sqrt{\frac{\pi}{3}} Y_1^0(\theta_p, \phi_p) - \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} \cdot 2 \sqrt{\frac{2\pi}{3}} Y_1^{-1}(\theta_p, \phi_p) \right. \\
&\quad \left. + \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i(\phi_1+\phi_2)} \cdot 2 \sqrt{\frac{2\pi}{3}} Y_1^1(\theta_p, \phi_p) + \sin \frac{\theta_1}{2} e^{-i\phi_1} \sin \frac{\theta_2}{2} \cdot 2 \sqrt{\frac{\pi}{3}} Y_1^0(\theta_p, \phi_p) \right]
\end{aligned}$$

$$\text{Applying orthogonality } \int_{\theta=0}^\pi \int_{\varphi=0}^{2\pi} Y_\ell^m Y_{\ell'}^{m'*} d\Omega = \delta_{\ell\ell'} \delta_{mm'},$$

(**MISTAKE** I forgot that orthogonality uses  $Y_1^{-1*}(\theta_p, \varphi_p)$  instead of  $Y_1^{-1}(\theta_p, \varphi_p)$ . Must add corresponding  $(-1)^m$  to each term )

$$\begin{aligned}
4\pi j_1(kr) \int_0^{2\pi} d\phi_p \int_0^\pi d\theta_p \sin \theta_p \Bigg[ & -2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_2} \\
& \times \underbrace{Y_1^0(\theta_p, \varphi_p) Y_1^{0*}(\theta_p, \varphi_p)}_{=1} Y_1^{0*}(\theta_r, \varphi_r) \\
& + (-2)\sqrt{\frac{2\pi}{3}} \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} \\
& \times \underbrace{Y_1^{-1}(\theta_p, \varphi_p) Y_1^{-1*}(\theta_p, \varphi_p)}_{=1} Y_1^{-1*}(\theta_r, \varphi_r) \\
& + 2\sqrt{\frac{2\pi}{3}} \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i(\phi_1+\phi_2)} \\
& \times \underbrace{Y_1^1(\theta_p, \varphi_p) Y_1^{1*}(\theta_p, \varphi_p)}_{=1} Y_1^{1*}(\theta_r, \varphi_r) \\
& + 2\sqrt{\frac{\pi}{3}} \sin \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{-i\phi_1} \\
& \times \underbrace{Y_1^0(\theta_p, \varphi_p) Y_1^{0*}(\theta_p, \varphi_p)}_{=1} Y_1^{0*}(\theta_r, \varphi_r) \Bigg]
\end{aligned}$$

## 5 Using the dot products

$$\begin{aligned}
\psi(\theta_r, \phi_r; \theta_1, \phi_1; \theta_2, \phi_2) \propto & \left[ \left( -2\sqrt{\frac{2\pi}{3}} \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} \right) Y_1^{*-1}(\theta_r, \phi_r) \right. \\
& + \left( -2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_2} + 2\sqrt{\frac{\pi}{3}} \sin \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{-i\phi_1} \right) Y_1^{*0}(\theta_r, \phi_r) \\
& \left. + \left( 2\sqrt{\frac{2\pi}{3}} \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i(\phi_1+\phi_2)} \right) Y_1^{*1}(\theta_r, \phi_r) \right]
\end{aligned}$$

$$\begin{aligned}
\text{old}(\theta_r, \phi_r; \theta_1, \phi_1; \theta_2, \phi_2) \propto & \left[ \left( -2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i\phi_2} \right) Y_1^{*-1}(\theta_r, \phi_r) \right. \\
& + \left( 2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} - 2\sqrt{\frac{\pi}{3}} \sin \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i(\phi_2-\phi_1)} \right) Y_1^{*0}(\theta_r, \phi_r) \\
& \left. + \left( 2\sqrt{\frac{2\pi}{3}} \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_1} \right) Y_1^{*1}(\theta_r, \phi_r) \right]
\end{aligned}$$

The goal is to express the wave function in a compact vector form. The wave function proportional to a sum over spherical harmonics could be written as a dot product involving vector  $\vec{C}$

$$\psi \propto j_1(pr) \left( \vec{C} \cdot \hat{r} \right)$$

Vector  $\vec{C}$  depends on the spin orientation of the fermions,  $\vec{s}_1$  and  $\vec{s}_2$ . We start with the components of  $\vec{C}$  derived from the matrix element calculation.

## 5.1 Initial expression for coefficients

In spherical basis

$$Y_1^0(\hat{r}) = \frac{3}{4\pi} n_z$$

$$Y_1^{\pm 1}(\hat{r}) = \mp \sqrt{\frac{3}{8\pi}} (n_x \pm i n_y)$$

where  $\hat{r} = (n_x, n_y, n_z)$

Rewriting expression as a dot product

$$\sum_m C_m Y_1^{m*} = \sqrt{\frac{3}{4\pi}} \vec{C} \cdot \vec{t}$$

$$\sum_m C_m Y_1^{m*} = A \cdot Y_1^{*-1} + B \cdot Y_1^{*0} + C \cdot Y_1^{*1} =$$

$$= A \left( -\sqrt{\frac{3}{4\pi}} (n_x - i n_y) \right) + B \left( -\sqrt{\frac{3}{4\pi}} n_z \right) + C \left( -\sqrt{\frac{3}{4\pi}} (n_x + i n_y) \right)$$

Group around  $\hat{r}$  components

$$\sum_m C_m Y_1^{m*} = \sqrt{\frac{3}{4\pi}} (-(A + C)n_x + i(A - C)n_y - Bn_z)$$

$$C_x = -(A + C) = 2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i\phi_2} - 2\sqrt{\frac{\pi}{3}} \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_1}$$

$$C_y = i(A - C) = -2i\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i\phi_2} - 2i\sqrt{\frac{2\pi}{3}} \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_1}$$

$$C_z = -B = -2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} + 2\sqrt{\frac{\pi}{3}} \sin \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i(\phi_2 - \phi_1)}$$

From the integral over the momentum direction of the decaying particles. we get the



following complex coefficients:

$$\begin{aligned} C_x &= 2\sqrt{\frac{\pi}{3}} (c_1 s_2 e^{i\phi_2} - s_1 c_2 e^{-i\phi_1}) \\ C_y &= -2i\sqrt{\frac{\pi}{3}} (c_1 s_2 e^{i\phi_2} + s_1 c_2 e^{-i\phi_1}) \\ C_z &= -2\sqrt{\frac{\pi}{3}} (c_1 c_2 - s_1 s_2 e^{i(\phi_2 - \phi_1)}) \end{aligned}$$

$$\begin{aligned} \vec{C} &= \alpha \vec{s}_1 + \beta \vec{s}_2 + \gamma (\vec{s}_1 \times \vec{s}_2) \\ C_x &= 2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i\phi_2} - 2\sqrt{\frac{\pi}{3}} \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_1} = \\ &= \alpha (\sin \theta_1 \cos \phi_1) + \beta (\sin \theta_2 \cos \phi_2) + \gamma (s_{1x}s_{2z} - s_{1z}s_{2y}) \\ &= \alpha (\sin \theta_1 \cos \phi_1) + \beta (\sin \theta_2 \cos \phi_2) + \gamma (\sin \theta_1 \cos \phi_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2 \sin \phi_2) \\ C_y &= -2i\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i\phi_2} - 2i\sqrt{\frac{2\pi}{3}} \sin \frac{\theta_1}{2} \cos \frac{\theta_2}{2} e^{-i\phi_1} = \\ &= \alpha (\sin \theta_1 \sin \phi_1) + \beta (\sin \theta_2 \sin \phi_2) + \gamma (s_{1z}s_{2x} - s_{1x}s_{2z}) \\ &= \alpha (\sin \theta_1 \sin \phi_1) + \beta (\sin \theta_2 \cos \phi_2) + \gamma (\cos \theta_1 \sin \theta_2 \cos \phi_2 - \sin \theta_1 \cos \phi_1 \cos \theta_2) \\ C_z &= -2\sqrt{\frac{\pi}{3}} \cos \frac{\theta_1}{2} \cos \frac{\theta_2}{2} + 2\sqrt{\frac{\pi}{3}} \sin \frac{\theta_1}{2} \sin \frac{\theta_2}{2} e^{i(\phi_2 - \phi_1)} = \\ &= \alpha \cos \theta_1 + \beta \cos \theta_2 + \gamma (s_{1x}s_{2y} - s_{1y}s_{2x}) \\ &= \alpha \cos \theta_1 + \beta \cos \theta_2 + \gamma (\sin \theta_1 \cos \phi_1 \sin \theta_2 \sin \phi_2 - \sin \theta_1 \sin \phi_1 \sin \theta_2 \cos \phi_2) \end{aligned}$$

Here,  $c_i = \cos(\theta_i/2)$  and  $s_i = \sin(\theta_i/2)$

## 5.2 Expressing coefficients using spin vector components

The components of the spin vectors are:

$$\vec{s}_i = (\sin \theta_i \cos \phi_i, \sin \theta_i \sin \phi_i, \cos \theta_i) = (s_{ix}, s_{iy}, s_{iz})$$

Some useful half-angle identities:

$$c_i^2 = \frac{1 + s_{iz}}{2}, \quad s_i^2 = \frac{1 - s_{iz}}{2}, \quad 2s_i c_i = \sqrt{1 - s_{iz}^2} = \sin \theta_i$$

Using these, we can express trigonometric functions of the angles:

$$\cos \phi_i = \frac{s_{ix}}{\sin \theta_i} = \frac{s_{ix}}{2s_i c_i}, \quad \sin \phi_i = \frac{s_{iy}}{\sin \theta_i} = \frac{s_{iy}}{2s_i c_i}$$

### 5.2.1 Separating real and imaginary parts

We expand the complex exponent  $e^{i\phi} = \cos \phi + i \sin \phi$  and separate the real and imaginary parts of each coefficient.

**z-components:** From equation (?), the imaginary part of  $C_z$  is:

$$\text{Im}(C_x) = 2\sqrt{\frac{\pi}{3}} s_1 s_2 \sin(\phi_2 - \phi_1)$$

$$\vec{s}_1 \times \vec{s}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ s_{1x} & s_{1y} & s_{1z} \\ s_{2x} & s_{2y} & s_{2z} \end{vmatrix} = \vec{i}(s_{1y}s_{2z} - s_{1z}s_{2y}) - \vec{j}(s_{1x}s_{2z} - s_{1z}s_{2x}) + \vec{k}(s_{1x}s_{2y} - s_{1y}s_{2x})$$

z-component of the cross product  $\vec{s}_1 \times \vec{s}_2$  is:

$$\begin{aligned} (\vec{s}_1 \times \vec{s}_2)_z &= s_{1x}s_{2y} - s_{1y}s_{2x} \\ &= (\sin \theta_1 \cos \phi_1)(\sin \theta_2 \sin \phi_2) - (\sin \theta_1 \sin \phi_1)(\sin \theta_2 \cos \phi_2) \\ &= \sin \theta_1 \sin \theta_2 (\cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2) \\ &= \sin \theta_1 \sin \theta_2 \sin(\phi_2 - \phi_1) \end{aligned}$$

Using the identity  $\sin \theta_i = 2s_i c_i$ , we get:

$$(\vec{s}_1 \times \vec{s}_2)_z = (2s_1 c_1)(2s_1 c_1) \sin(\phi_2 - \phi_1)$$

Now, we can express term form  $\text{Im}(C_z)$ :

$$s_1 s_2 \sin(\phi_2 - \phi_1) = \frac{(\vec{s}_1 \times \vec{s}_2)_z}{4c_1 c_2}$$

Substitute back into the  $\text{Im}(C_z)$ :

$$\text{Im}(C_z) = 2\sqrt{\frac{\pi}{3}} \frac{(\vec{s}_1 \times \vec{s}_2)_z}{4c_1 c_2} = \frac{\sqrt{\pi/3}}{2c_1 c_2} (\vec{s}_1 \times \vec{s}_2)_z$$

**x- and y-components:** Following a similar procedure for  $C_x$  and  $C_y$ :

$$\text{Im}(C_x) = 2\sqrt{\frac{\pi}{3}} (c_1 s_2 \sin \phi_2 + s_1 c_2 \sin \phi_1) = \frac{\sqrt{\pi/3}}{c_1 c_2} (c_1^2 s_{2y} + c_2^2 s_{1y})$$

$$\text{Im}(C_y) = -\text{Re} \left( 2\sqrt{\frac{\pi}{3}} (c_1 s_2 e^{i\phi_2} + s_1 c_2 e^{-i\phi_1}) \right) = -2\sqrt{\frac{\pi}{3}} (c_1 s_2 \cos \phi_2 + s_1 c_2 \cos \phi_1) = -\frac{\sqrt{\pi/3}}{c_1 c_2} (c_1^2 s_{2x} + c_2^2 s_{1x})$$

Replacing  $c_i^2 = (1 + s_{iz})/2$ , we get:

$$\begin{aligned}\text{Im}(C_x) &= \frac{\sqrt{\pi/3}}{2c_1c_2} [(1 + s_{1z})s_{2y} + (1 + s_{2z})s_{1y}] \\ \text{Im}(C_y) &= -\frac{\sqrt{\pi/3}}{2c_1c_2} [(1 + s_{1z})s_{2x} + (1 + s_{2z})s_{1x}]\end{aligned}$$

The terms in the brackets can be expanded:

$$\begin{aligned}(1 + s_{1z})s_{2y} + (1 + s_{2z})s_{1y} &= s_{2y} + s_{1y} + s_{1z}s_{2y} + s_{1y}s_{2z} \\ &= s_{1y} + s_{2y} + (s_{1y}s_{2z} - s_{1z}s_{2y}) + 2s_{1z}s_{2y} \\ &= s_{1y} + s_{2y} + (\vec{s}_1 \times \vec{s}_2)_x + 2s_{1z}s_{2y} \\ (1 + s_{1z})s_{2x} + (1 + s_{2z})s_{1x} &= s_{2x} + s_{1x} + s_{1z}s_{2x} + s_{2z}s_{1x} \\ &= s_{1x} + s_{2x} + (s_{1z}s_{2x} - s_{1x}s_{2z}) + 2s_{1x}s_{2z} \\ &= s_{1x} + s_{2x} + (\vec{s}_1 \times \vec{s}_2)_y + 2s_{1x}s_{2z}\end{aligned}$$

### 5.2.2 Analysis of $\ell = 1$ contributions

The wave function's dependence on the direction  $\hat{r}$  comes from terms with  $\ell = 1$ , which behave like vectors. All terms derived from the expansion are vectorial and contribute to  $\vec{C}_{\ell=1}$ . There are no scalar ( $\ell = 0$ ) contributions like  $\vec{s}_1 \cdot \vec{s}_2$ .

The components of the imaginary part of  $\vec{C}$  can be grouped into distinct physical contributions:

$$\begin{aligned}\text{Im}(C_x) &= \frac{\sqrt{\pi/3}}{2c_1c_2} \left[ \underbrace{(s_{1y} + s_{2y})}_{\text{Sum of spins}} + \underbrace{(\vec{s}_1 \times \vec{s}_2)_x}_{\text{Spin correlation}} + \underbrace{2s_{1z}s_{2y}}_{\text{Higher-order}} \right] \\ \text{Im}(C_y) &= -\frac{\sqrt{\pi/3}}{2c_1c_2} \left[ \underbrace{(s_{1x} + s_{2x})}_{\text{Sum of spins}} + \underbrace{(\vec{s}_1 \times \vec{s}_2)_y}_{\text{Spin correlation}} + \underbrace{2s_{1x}s_{2z}}_{\text{Higher-order}} \right] \\ \text{Im}(C_z) &= \frac{\sqrt{\pi/3}}{2c_1c_2} \left[ \underbrace{(\vec{s}_1 \times \vec{s}_2)_z}_{\text{Spin correlation}} \right]\end{aligned}$$

The complete expression for the imaginary part of  $\vec{C}$  is:

$$\begin{aligned}\text{Im}(C_x) &= \frac{\sqrt{\pi/3}}{2c_1c_2} [(s_{1y} + s_{2y}) + (\vec{s}_1 \times \vec{s}_2)_x + 2s_{1z}s_{2y}] \\ \text{Im}(C_y) &= -\frac{\sqrt{\pi/3}}{2c_1c_2} [(s_{1x} + s_{2x}) + (\vec{s}_1 \times \vec{s}_2)_y + 2s_{1x}s_{2z}] \\ \text{Im}(C_z) &= \frac{\sqrt{\pi/3}}{2c_1c_2} (\vec{s}_1 \times \vec{s}_2)_z\end{aligned}$$

## 6 Conclusion